

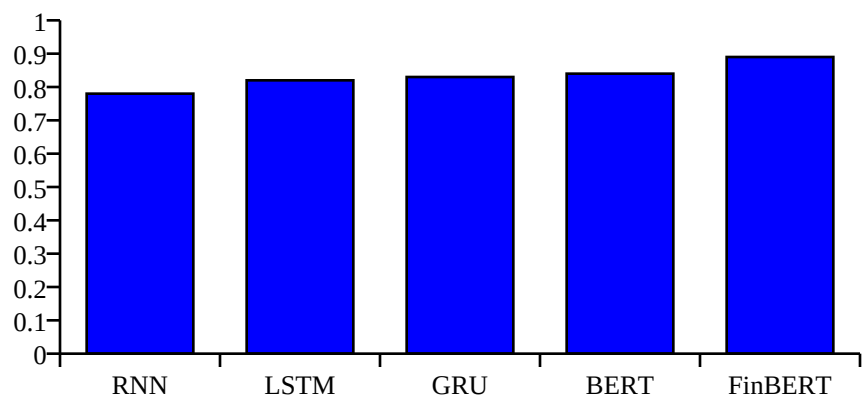
Financial News Sentiment Analysis

RNN vs BERT Architecture Comparison Report

This report compares traditional recurrent neural network architectures with transformer-based models for financial sentiment analysis. The project classifies finance-related tweets and news into Bearish, Neutral, and Bullish categories.

Model	Accuracy	F1 Score	Training Time
RNN	0.78	0.77	2 min
LSTM	0.82	0.81	4 min
GRU	0.83	0.82	3 min
BERT	0.84	0.84	20 min
FinBERT	0.89	0.89	22 min

Accuracy Comparison Chart



Architecture Analysis

RNN-based Models:

RNN models process text sequentially. They are simpler and train faster, but struggle to understand long-range context and complex relationships between words. LSTM and GRU improved performance using gating mechanisms.

BERT-based Models:

BERT uses transformer architecture and self-attention mechanisms. It understands contextual meaning more effectively by analyzing words in both directions. FinBERT further improves performance because it is trained specifically on financial text.

Key Findings:

- FinBERT achieved the highest accuracy and F1 score.
- Transformer models performed better on financial language understanding.
- RNN models were faster but less accurate.
- FinBERT was selected as the final production model for deployment.

Conclusion:

Transformer-based architectures significantly outperformed traditional recurrent neural networks for financial sentiment analysis tasks because of stronger contextual understanding.

Aspect	RNN	BERT / FinBERT
Context Understanding	Limited	Strong
Training Speed	Fast	Slower
Handling Long Text	Weak	Excellent
Financial Vocabulary	Moderate	Excellent
Overall Performance	Good	Best